

PROJECT DOCUMENTATION

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Abstract

This project analyzes a housing dataset using Tableau Public to identify trends in sale prices and housing characteristics. Interactive dashboards and stories were created and embedded into a Flask web application.

1. Introduction

The housing market produces large volumes of data. Tableau enables decision-makers to explore this data visually and discover patterns that are difficult to identify from spreadsheets.

2. Problem Statement

Develop an interactive Tableau dashboard and story to analyze housing prices, bedrooms, bathrooms, living area, floors, and geographical distribution, then integrate the dashboard into a Flask web application.

3. Objectives

- Analyze housing data.
- Build interactive Tableau dashboards.
- Create a Tableau Story.
- Publish on Tableau Public.
- Integrate the dashboard using Flask.
- Upload project to GitHub.

4. Software & Tools

Python, Flask, HTML, Tableau Public, GitHub, Visual Studio Code.

5. Dataset Description

Dataset fields include Sale Price, Bedrooms, Bathrooms, Living Area, Floors, Zipcode, Renovation information, and House Location.

6. Methodology

Data Collection → Data Cleaning → Data Visualization → Dashboard Design → Story Creation → Tableau Public Publishing → Flask Integration → GitHub Repository.

7. Visualizations

Scatter Plot (Price vs Living Area), Map (House Locations), Bar Chart (Average Price by Bedrooms), Histogram (Bathrooms Distribution), Pie Chart (Houses by Floors).

8. Dashboard

An interactive dashboard combines all charts with filters for better exploration of housing market trends.

9. Storyboard

A Tableau Story presents the project in sequential steps, explaining each visualization and the business insights.

10. Flask Integration

A Flask application renders index.html and embeds the published Tableau dashboard using an iframe.

11. GitHub Repository

The repository contains app.py, templates/index.html and requirements.txt.

12. Testing

Verified successful loading of the dashboard, responsive visualization rendering, working filters and successful web embedding.

13. Results

The analysis shows that larger living areas generally have higher sale prices. Homes with more bedrooms tend to have higher average prices. Most houses contain 2–3 bathrooms, while one-floor houses are the most common.

14. Conclusion

The project successfully demonstrates interactive business intelligence using Tableau Public and web integration through Flask.

15. Future Scope

Add predictive analytics, machine learning models, real-time datasets and advanced filtering.

Project Completion Summary

Module	Status
Data Collection	Completed
Data Preparation	Completed
Visualization	Completed
Dashboard	Completed
Storyboard	Completed
Flask Integration	Completed
GitHub Repository	Completed
Testing	Completed

References

1. Tableau Public Documentation

2. Flask Documentation
3. Python Documentation
4. GitHub Documentation